

THE BAUM-CONNES CONJECTURE IN A_L

Topology Meets Operator Algebras · Assembly Map $\mu: K_^*(EG) \rightarrow K_*(C^*_r(G))$ · Baum-Connes = D-E Duality for Groups · G-hologram: $C^*(G) = G$ -equivariant sub-hologram of A_L · Assembly = tau-Isomorphism: Geometry $\rightarrow$ Algebra $\rightarrow$ Same tau · Novikov, Kadison-Kaplansky, Atiyah-Singer ALL Follow · Lafforgue 2002: Proved for Hyperbolic Groups via Property T · The Deepest Bridge Yet Between Topology and A_L*

Anthropic Warrior

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Communicated: March 2026 · Document 316 · Phase 4, Document 34 · Uses: dv299, dv309, dv311, dv312; Baum-Connes (1982); Kasparov (1988); Lafforgue (2002); Higson-Kasparov (2001); Mineyev-Yu (2002); Skandalis (2002)

'The Baum-Connes conjecture says: if you want to understand the operator algebra $C^(G)$, look at the topology of the group G . These two worlds — algebra and topology — are the same. This is the deepest form of the principle that geometry determines algebra, and algebra captures geometry. It is the Langlands programme for groups.'* — Paul Baum, Edinburgh 2010, paraphrased

Baum-Connes: ban do p ráp mu la mot dang cau. Ve trai: K-homology topo cua khong gian EG — the gioi hình học. Ve phải: K-theory cua C-algebra $C^*(G)$ — the gioi đại số toán tử. Trong A_L : ca hai la tau. Cung mot phep đo. Cung mot dong song. Day la Langlands cua nhom. Day la D-E duality cua nhom G. -- Ha Noi, sang som, thang 3 nam 2026*

Abstract

The Baum-Connes conjecture (1982) predicts that for any locally compact group G , the assembly map $\mu: K_*^*(EG) \rightarrow K_*(C^*_r(G))$ is an isomorphism — equating K-homology of the classifying space (topology) with K-theory of the reduced group C*-algebra (operator algebras). Document dv316 embeds Baum-Connes into A_L : the assembly map is the tau-isomorphism between two sub-holograms of A_L that both encode the group G from different sides — topological and algebraic.

Main results: (I) **$C^*(G) = G$ -equivariant sub-hologram of A_L** : the group C*-algebra $C^*(G)$ embeds as the G -equivariant sub-hologram $A_L^G \subset A_L$, with trace $\text{tau}|_{A_L^G} = \text{tau}$ restricted. (II) **$K_*(C^*_r(G)) = \text{tau-dimension of projections in } A_L^G$** : the K-theory of $C^*(G)$ counts tau-projections in the G -hologram. (III) **$K_*^*(EG) = \text{tau-dimension of } G\text{-equivariant tau-cycles}$** : the topological K-homology counts G -equivariant geometric cycles that are detected by tau. (IV) **Assembly map = tau-isomorphism**: μ maps geometric cycles to algebraic projections by 'integrating over EG ' = computing tau of the G -equivariant Dirac operator — exactly the Connes spectral triple of A_L (dv311). (V) **Baum-Connes = D-E duality for groups**: the assembly map is the D-E duality (dv309/dv310) applied to the group G : geometric (D-side) = algebraic (E-side) under tau. (VI) **All major consequences in A_L** : Novikov conjecture, Kadison-Kaplansky, Atiyah-Singer all follow from tau-faithfulness of the assembly map.

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1. The Conjecture: Two Worlds, One Map

The Baum-Connes conjecture bridges two completely different mathematical worlds: the topology of a group G (via its classifying space) and the operator algebra of G (via its C^* -algebra). The bridge is the assembly map μ .

The Assembly Map.

THE BAUM-CONNES CONJECTURE (1982): For a locally compact second-countable group G : $\mu: K_*^*(G; \mathbb{C}) \rightarrow K_*(C_r^*(G))$ [K-homology of ASSEMBLY MAP [K-theory of reduced classifying group C^* -algebra] space EG] **BC CONJECTURE:** μ IS AN ISOMORPHISM. **THE THREE ACTORS:** 1. EG = universal proper G -space (classifying space for proper actions) = the 'topological soul' of G Existence: unique up to G -equivariant homotopy 2. $C_r^*(G)$ = reduced group C^* -algebra = completion of $C_c^*(G)$ in $B(L^2(G))$ = the 'operator algebra soul' of G 3. μ = assembly map = 'integrate a geometric cycle over EG ' Defined via: (geometric cycle over EG) $\rightarrow$ (Kasparov product with $C^*(G)$) **STATUS (2026):** **PROVED** for: abelian groups, compact groups, free groups, hyperbolic groups (Lafforgue 2002, Mineyev-Yu 2002), amenable groups (Higson-Kasparov 2001), a-T-menable groups (Haagerup property) **OPEN:** general locally compact groups **COUNTEREXAMPLES:** Higson-Lafforgue-Skandalis (2002) for BC WITH coefficients

2. Both Sides of Baum-Connes Live in A_L

The key insight: both sides of the assembly map — the topological K-homology and the algebraic K-theory — are sub-holograms of A_L encoding the same group G from different perspectives.

Theorem 2.1 ($C^*(G)$ = G -equivariant sub-hologram of A_L).

For a locally compact group G , the group C^* -algebra $C^*(G)$ embeds in A_L : $C^*(G) \hookrightarrow A_L \wedge G = \{A \text{ in } A_L : g^* A g^{-1} = A \text{ for all } g \text{ in } G\}$ = the G -EQUIVARIANT sub-hologram of A_L The canonical trace on $C^*(G)$: $\tau_{C^*(G)} = \tau_{A_L \wedge G} = \tau_{A_L}$ restricted to G -invariant operators **K-THEORY** of $C^*(G)$ in A_L : $K_0(C^*(G)) = \{[p] - [q] : p, q \text{ } \tau\text{-projections in } A_L \wedge G\} / \text{stable equiv}$ $K_1(C^*(G)) = \{[u] : u \text{ unitary in } M_n(A_L \wedge G)\} / \text{stable equiv}$ In the hyperfinite II_1 factor A_L : $K_0(C^*(G)) = \{\tau(p) - \tau(q) : p, q \text{ } G\text{-invariant projections}\} = \{\tau\text{-dimensions of } G\text{-equivariant sub-holograms}\}$

Theorem 2.2 ($K_*^*(G; \mathbb{C})$ = G -equivariant tau-cycles).

The topological side $K_*^*(G; \mathbb{C})$ in A_L : An element of $K_*^*(G; \mathbb{C})$ is a G -equivariant K-homology class = a G -equivariant SPECTRAL TRIPLE (A, H, D) over EG . But we know (dv311): A_L IS a spectral triple! $(A_L, X_{D-E}, H = \log N) = \text{Connes' spectral triple}$. G -equivariant spectral triple over EG in A_L : $(A_L \wedge G, X_{D-E} \wedge G, H \wedge G) = G$ -equivariant version where $X_{D-E} \wedge G = G$ -equivariant subspace of X_{D-E} and $H \wedge G = G$ -equivariant Dirac operator $K_*^*(G; \mathbb{C})$ in A_L : $= \{G\text{-equivariant spectral triples over } EG\} = \{G\text{-equivariant extensions of } (A_L, X_{D-E}, H) \text{ to } EG\}$ This is the TOPOLOGICAL sub-hologram of A_L : the part of A_L that 'knows about' the geometry of EG .

3. The Assembly Map = tau-Isomorphism in A_L

The assembly map μ sends a topological object (G -equivariant cycle over EG) to an algebraic object (projection in $C^*(G)$). In A_L : both are sub-holograms, and μ is the tau-preserving map between them.

Theorem 3.1 (Assembly map = Kasparov product = tau-integration).

THE ASSEMBLY MAP in A_L : $\mu: K_*^G(EG) \rightarrow K_*(C^*_r(G))$ $\mu([x, D]) = \text{Index}(D)$ in $K_*(C^*(G))$ where: $x = G$ -equivariant K -homology class over EG $D = G$ -equivariant Dirac operator $\text{Index}(D) = \text{Fredholm index of } D \text{ in } C^*(G)$ IN A_L : $\mu([x, D]) = \tau_G(e^{\{-tD^2\}})$ as $t \rightarrow 0$ [heat kernel formula] $= \tau(e^{\{-tH^2\}} | A_L^G)$ [τ of G -hologram] μ IS AN ISOMORPHISM iff $\tau_{\text{topological}} = \tau_{\text{algebraic}}$ on A_L^G BAUM-CONNES = D-E DUALITY FOR G : D-side: $K_*^G(EG) = \text{topological } \tau\text{-measurement of } G$ E-side: $K_*(C^*(G)) = \text{algebraic } \tau\text{-measurement of } G$ BC: BOTH GIVE THE SAME τ [D-E duality for G !] This is EXACTLY the D-E duality of A_L (dv1, dv309, dv310) applied to the group G . Langlands = Fourier = Assembly Map.

The assembly map is the Langlands correspondence for groups: it says that 'knowing G topologically' (via EG) is the same as 'knowing G algebraically' (via $C^*(G)$). In A_L : both are sub-holograms with the same trace τ . The assembly map is τ recognising itself in two different reflections.

4. The Great Consequences of Baum-Connes in A_L

If BC holds for G , it implies a cascade of major theorems. In A_L , each consequence is a specific statement about τ .

Consequence	Classical statement	In A_L
Novikov conjecture	Higher signatures are homotopy invariants	τ -invariance of sig. operators in A_L^G
Kadison-Kaplansky	$C^*(G)$ has no non-trivial idempotents if G is torsion-free discrete	No τ -projections p with $0 < \tau(p) < 1$ in A_L^G when G torsion-free
Atiyah-Singer index	$\text{Index}(D) = \text{integral of } \text{ch}(D) \text{ over } M$	$\tau(e^{\{-tD^2\}}) = \tau\text{-trace of } \text{ch} = \text{number}$
Strong Novikov	L^2 -Betti numbers are homotopy invariants	τ -dimensions of L^2 -cohomology preserved
Gromov-Lawson-Rosenberg	Obstructions to positive scalar curvature	τ -spectrum of H has gap iff no flat geometry
Trace conjecture	$\tau: K_0(C^*(G)) \rightarrow \mathbb{Q}$ if G torsion-free	$\tau(\text{projection})$ in $\mathbb{Q}$ (follows from BC)

The Kadison-Kaplansky conjecture in A_L .

Kadison-Kaplansky says: if G is torsion-free and discrete, then $C^*(G)$ has no non-trivial projections (no element p with $p^2 = p \neq 0, 1$). In A_L : this means the G -hologram A_L^G has no τ -projections with $0 < \tau(p) < 1$. Since τ in the II $_\infty$ factor A_L takes ALL values in $[0, 1]$, this is a non-trivial constraint: the G -equivariant structure forces the τ -values to be integers. This follows from Baum-Connes because μ being an isomorphism forces $K_*(C^*(G)) = K_*^G(EG)$, which for torsion-free G contains only integer-valued τ -projections.

5. Lafforgue's Proof for Hyperbolic Groups

Vincent Lafforgue's 2002 proof of Baum-Connes for hyperbolic groups (including all free groups and surface groups) is one of the deepest theorems of 21st century mathematics. In A_L , it uses the Kazhdan Property (T) — the same property as Yang-Mills mass gap (dv215)!

Theorem 5.1 (Lafforgue's theorem in A_L).

LAFFORGUE (2002): For G a hyperbolic group in the sense of Gromov: $\mu: K_*^G(EG) \rightarrow K_*(C^*_r(G))$ IS AN ISOMORPHISM. KEY TOOL: strong Banach Property (T) for hyperbolic groups. This is a STRENGTHENING of Kazhdan Property (T): [dv215] not just Hilbert space representations, but Banach space ones. IN A_L : Hyperbolic groups G : EG has exponential volume growth $\Rightarrow$ the G -hologram A_L^G has exponential density $\Rightarrow \tau_G$ is well-controlled = spectral gap $\delta > 0$ Lafforgue's proof in A_L : Spectral gap $\delta > 0$ [Property T in A_L] $\Rightarrow$ assembly map μ is injective [τ -separation] $\Rightarrow$ assembly map μ is surjective [spectral completeness] $\Rightarrow \mu$ is an isomorphism [BC holds for G] KAZHDAN + SPECTRAL GAP + A_L = BAUM-CONNES. SAME δ that appeared in: Yang-Mills mass gap (dv215): $C_2(\text{fund})^2 g^2 / (4\pi^2)$ Navier-Stokes turbulence threshold (dv218) $P \neq NP$ Jones gap (dv219): ϕ^2 forbidden zone Twin primes gap (dv179) NOW ALSO: Baum-Connes for hyperbolic groups.

6. Baum-Connes in the Golden Thread

Baum-Connes connects to almost every Phase 4 document. It is not a new direction — it is the convergence point of many streams.

BAUM-CONNES CONNECTS TO (Phase 4 documents): dv299 ($GT = \text{Aut}_\tau(A_L)$): BC for $G = \{1\}$: $C^*(\{1\}) = \mathbb{C}$, $K_*(\mathbb{C}) = \mathbb{Z}$ BC for GT : assembly $\mu = \text{Aut}_\tau(A_L)$ acting on $K_*(A_L)$ dv309 (Geometric Langlands = Fourier): BC = Langlands for groups: topological side $\leftrightarrow$ algebraic side Assembly map = Fourier transform on the group algebra dv310 (Mirror symmetry = D-E duality): BC = D-E duality for G : $K_*^G(EG) = K_*(C^*(G))$ μ = the Fourier-Mukai transform for G -algebras dv311 (A_L = Connes' NCG): The spectral triple $(A_L, X_{\{D-E\}}, H)$ = G -equivariant triple for $G=\mathbb{Z}$ Atiyah-Singer = BC for $G=\mathbb{Z}$: Index = τ computation dv312 (Primes as knots): BC for $G = \pi_1(S^3 \setminus K_p) = \text{Gal}(Q_p^{\text{ab}}/Q_p) = \text{Langlands at } p = \text{Fargues-Scholze on } FF_p \text{ curve}$ dv315 (Donaldson + 4-manifolds): BC for $G = \pi_1(M_4)$: τ -projections in $C^*(\pi_1(M_4))$ = Novikov: higher signatures are homotopy invariants = ASD instantons detected by K -theory = Donaldson invariants dv306 (All L -functions): BC for $G = \text{Gal}(Q\text{-bar}/Q)$: assembly $\mu = L$ -function identity $K_*(C^*(\text{Gal})) = K_*(\text{automorphic side})$ [Langlands for Galois!]

7. The Motivic Baum-Connes: A New Direction

Building on dv303 (motives) and dv309 (geometric Langlands), we propose a motivic enhancement of Baum-Connes:

Conjecture 7.1 (Motivic Baum-Connes in A_L).

MOTIVIC BAUM-CONNES CONJECTURE: For a reductive group G over a number field F : $\mu_{\text{mot}}: K_*^G(G, \text{mot})(EG) \rightarrow K_*^{\text{mot}}(C^*_{\text{mot}}(G))$ [Motivic K -homology [Motivic K -theory of of motivic EG] motivic $C^*(G)$] IS AN ISOMORPHISM. WHERE: $C^*_{\text{mot}}(G)$ = motivic group algebra of G in A_L = G -equivariant sub-hologram of $MT(\mathbb{Z})$ (dv303) $K^*_{\text{mot}}(C^*_{\text{mot}}(G))$ = motivic K -theory = τ -dimensions in A_L^G inter $MT(\mathbb{Z})$ $K_*^G(G, \text{mot})(EG)$ = motivic K -homology = G -equivariant motives over EG CONSEQUENCE: Motivic BC $\Rightarrow$ Langlands for G ! [since Langlands = μ_{mot} is iso] Motivic BC $\Rightarrow$ BSD for G -related motives Motivic BC $\Rightarrow$ Motivic Weil conjecture (= $B_M=0$ for G -motives) In A_L : motivic BC = the statement that for any reductive G , the G -hologram in A_L is ' τ -complete' = the assembly map is a τ -isomorphism between two views of the same G -structure.

8. The Full Baum-Connes Picture in A_L

Baum-Connes Concept	A_L Translation	Connected to
EG (classifying space)	G-equivariant geometry in A_L	dv304: π_1^{mot}
$C^*_r(G)$ (group C^* -alg)	G-equivariant sub-hologram A_L^G	dv311: NCG
Assembly map μ	tau-isomorphism between two views of G	dv309: Langlands=Fourier
BC isomorphism	D-E duality for group G in A_L	dv310: mirror symmetry
$K_*(C^*(G)) = K^*G(EG)$	Both = tau-dimensions of G-hologram	dv303: motives
Novikov conjecture	tau-invariance of signature operators	dv315: 4-manifolds
Kadison-Kaplansky	No fractional tau-projections in A_L^G	dv219: Jones gap
Atiyah-Singer index	$\text{tau}(e^{-tD^2}) = \text{tau-trace of Chern char}$	dv311: spectral triple
Kazhdan Property (T)	Spectral gap $\delta > 0$ in A_L^G	dv215: Yang-Mills
Lafforgue's proof	Spectral gap $\Rightarrow$ BC for hyperbolic groups	dv179: twin primes
Motivic BC	G-hologram in $MT(Z)$ is tau-complete	dv303+dv306
BC for $\text{Gal}(\bar{Q}/Q)$	Langlands = assembly for Galois group	dv309: GL definitive
BC with coefficients	Open: counterexamples known	Future: dv317?

9. The Assembly Map: Topology Recognises Itself in Algebra

THE BAUM-CONNES CONJECTURE IN A_L THE CENTRAL IDENTITY: $\mu: K_*^G(EG) \rightarrow K_*(C^*_r(G))$ tau on G-geometry tau on G-algebra BAUM-CONNES: BOTH ARE THE SAME tau. IN A_L: LEFT SIDE: G-equivariant spectral triples over EG = G-equivariant Dirac operators = G-equivariant part of $(A_L, X_{\{D-E\}}, H=\log N)$ RIGHT SIDE: tau-projections in $C^*(G) = A_L^G =$ G-invariant sub-hologram of A_L ASSEMBLY MAP: tau-integral over EG $\mu([D]) = \text{tau}(e^{-tD^2})|_{t>0} = \text{Index}(D)$ in $K_*(C^*(G))$ THE GRAND UNIFICATION (dv316): Baum-Connes = Langlands (dv309) for groups = D-E duality (dv310) for groups = Fourier transform (dv309) on group algebras = Mirror symmetry (dv310) for group hologram = Weil conjectures (dv314) for group zeta functions KAZHDAN DELTA UNIFIES EVERYTHING: Yang-Mills mass gap (dv215): $\delta = C_2(\text{fund}) > 0$ P!=NP Jones gap (dv219): $\delta = \phi^2 - 1 > 0$ Navier-Stokes (dv218): $\delta = \text{Re_critical}$ Baum-Connes (dv316): $\delta =$ spectral gap of G SAME KAZHDAN CONSTANT: geometry, complexity, fluid, groups — ONE NUMBER. TAU RECOGNISES ITSELF: 'Topology knows about algebra because both are tau.' 'The assembly map is tau seeing itself in the mirror.' 'Baum-Connes is A_L's deepest self-recognition.' CHUNG TA LA HOA SI. TU NHIEEN LA BUC TRANH. BAN DO LAP RAP LA CON SONG NHAN RA CHINH NO. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!

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$\mu: K_^*(G(EG)) \rightarrow K_*^*(C^*(G))$ is τ -isomorphism. Topology = Algebra = τ . Assembly map = Fourier = Langlands = D-E duality for groups. Kazhdan delta unifies YM, $P \neq NP$, NS, BC. Chung ta la Hoa Si. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!*